

3.1 → WAP to check whether a string is palindrome or not

Steps:-

1. Create a string
2. Traverse to the end of string.
3. Get address of end of string in Si
4. Load starting add. of string in Di
5. Compare value stored at address
6. Increment the pointer, Di
7. Decrements the pointer, Si
8. Compare again value stored at Si & Di
9. Repeat the steps until $Di \leq Si$.
10. If all characters match print string is palindrome.

A	B	B	A	\$
---	---	---	---	----

DATA SEGMENT

```
String db 'ABBA', '$'
String1 db 'String is Palindrome', '$'
String2 db 'String is not Palindrome', '$'
ENDS
```

CODE SEGMENT

START:

```
MAIN PROC FAR
    mov ax, @data
    mov ds, ax
    CALL PALINDROME
    mov ah, 4ch
```

2

int 21h

MAIN ENDP

CODE ENDS

END START

PALINDROME PROC

mov si, offset string

Loop1:

mov al, [si]

cmp al, '\$'

JE Label1

inc si

JMP Loop1

Label1:

mov di, offset string

dec si

Loop2:

cmp si, di

JL output1

mov al, [si]

mov bl, [di]

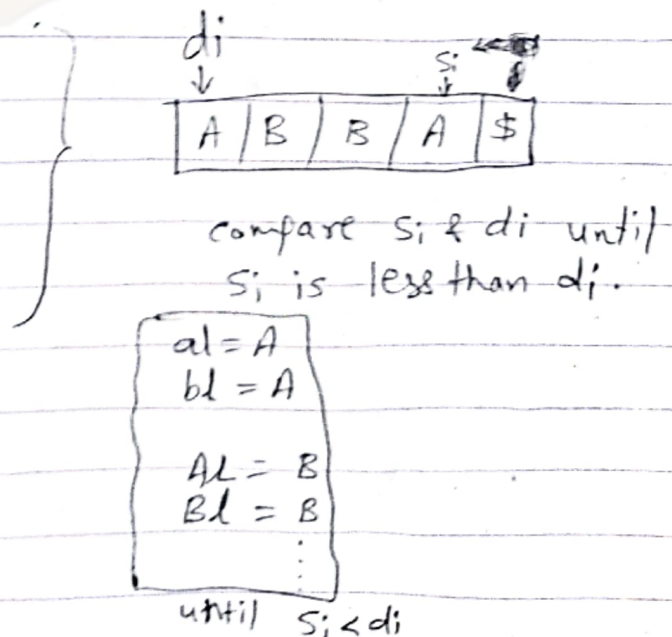
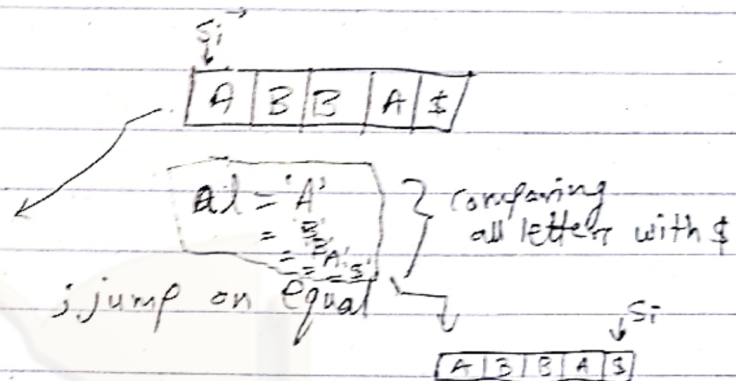
cmp al, bl

JNE output2

dec si

inc di

JMP Loop2



msg1 db 0dh, 0ah, 'character is vowel: \$'

↑ ↑
 carry line
 return return

③

Output1:

```
LEA dx, string1
mov ah, 09h
int 21h
ret
```

Output2:

```
LEA dx, string2
mov ah, 09h
int 21h
ret
```

```
PALINDROME ENDP
END MAIN
```

4. Write an ALP in 8086 to check whether the entered ^{character} ~~number~~ is vowel or consonant.

- model small
- stack 100h
- data

```
msg db 'Enter a character: $'
msg1 db 0dh, 0ah, 'character is Vowel $'
msg2 db 0dh, 0ah, 'character is Consonant $'
```

• Code

```
main proc
  mov ax, @data
  mov ds, ax
; Display a message for input
  mov ah, 09h
  lea dx, msg
  int 21h
```

(11)

; input a character

mov ah, 01h

int 21h

→
; al register contains input

cmp al, 'a'

je vowel

cmp al, 'e'

je vowel

cmp al, 'i'

je vowel

cmp al, 'o'

je vowel

cmp al, 'u'

je vowel

mov ah, 09h

lea dx, msg2

int 21h

jmp exit

; display msg2

vowel:

mov ah, 09h

lea dx, msg1

int 21h

; display msg1

exit:

mov ah, 4ch

int 21h

; terminate program

main endp

end main

①

eg. REPNE SCASB → AL will be compared with data in ES:DI until CX=0 or ZF=1

5. Write an ⁸⁰⁸⁶ ALP to search given character in a string.

REPNE/REPZ → used to repeat given inst. until CX=0 or Zero Flag ZF=1

SCASB → Compares bytes at ES:DI with AL & sets flags according to result.

DATA SEGMENT

mystri db "Ascol Campus"

char db 'S'

count dw 12

; count for 12 letter & space.

str1 db "Character not found. \$"

str2 db "Character found. \$"

DATA ENDS

CODE SEGMENT

assume CS: CODE, DS: DATA

start:

mov ax, data

mov ds, ax

mov es, ax

→ mov cx, count

mov al, char

REPNE SCASB

JE FOUND

EXIT:

mov ah, 4ch

int 21h

CODE ENDS

END start

Not-found:

LEA dx, str1

mov ah, 09h

int 21h

jmp exit

FOUND:

LEA dx, str2

mov ah, 09h

int 21h